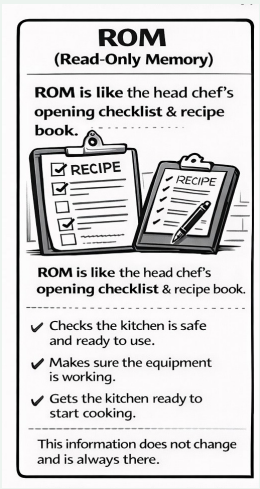
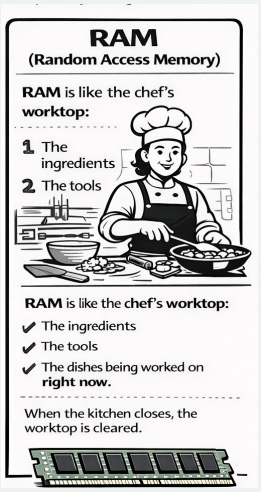
01 · THEORY

Primary memory

The chef’s kitchen helps us remember what RAM, ROM and virtual memory do.



ROM · opening instructions



RAM · current work

Three things this guide will make clear

01 Start-up instructions

ROM keeps the instructions needed to get the computer started.

02 Working memory

RAM holds programs and data that are being used right now.

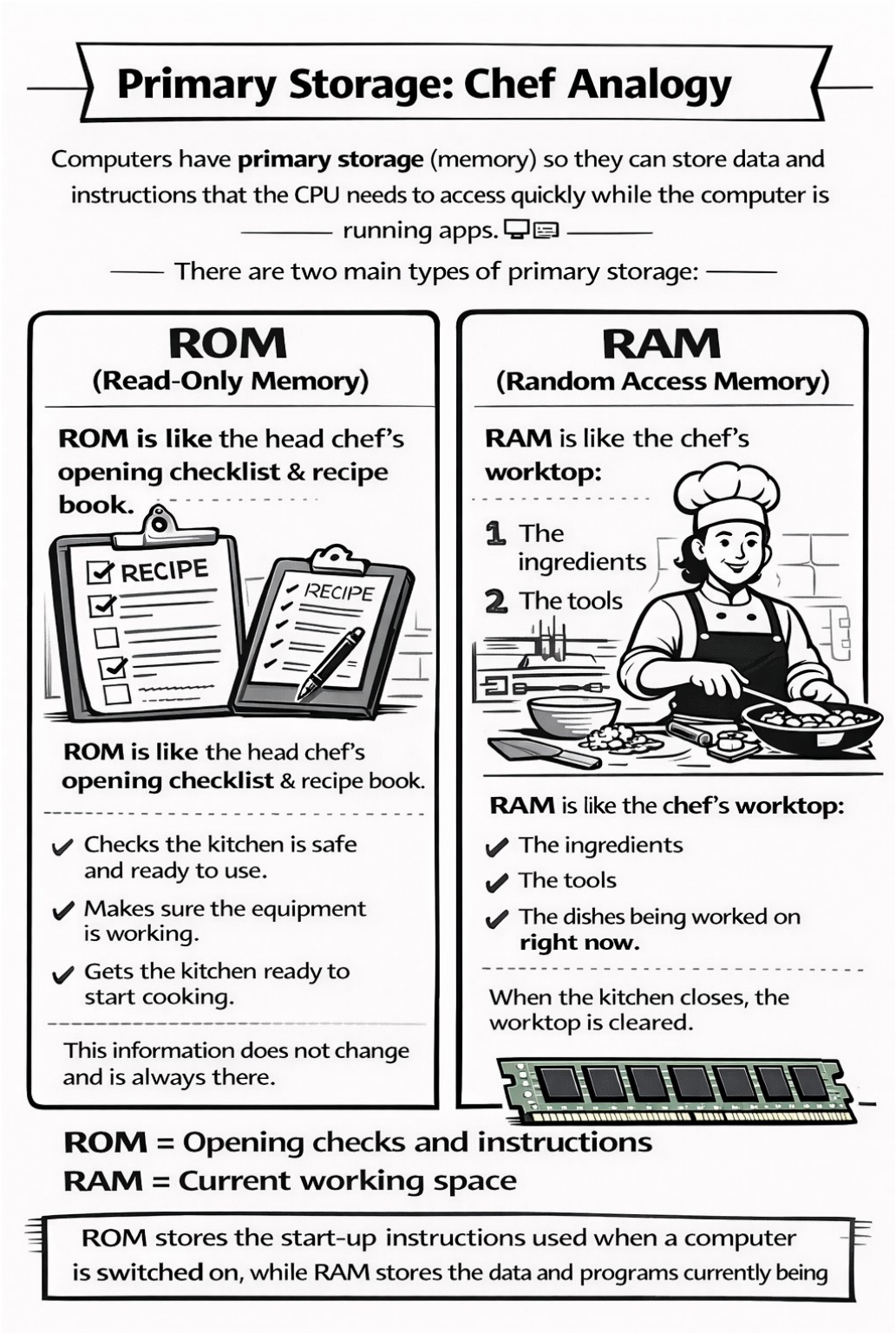
03 When RAM is full

Virtual memory temporarily uses slower secondary storage.

01 · THEORY

RAM and ROM in the kitchen

The illustration compares an opening checklist with a worktop.



01 · THEORY

From kitchen to examination

Use the visual analogy to understand the idea; use the precise technical words when answering.

ROM

Opening checklist in the analogy. In a computer, ROM is non-volatile memory that stores start-up instructions (firmware). The computer can read these instructions when it powers on.

Exam wording: ROM stores the start-up instructions and retains its contents when power is removed.

RAM

Chef’s worktop in the analogy. RAM stores the instructions and data for programs currently running. It can be read and written to, but its contents are lost when the computer powers off.

Exam wording: RAM is volatile, read/write memory for programs and data in current use.

Compare them — the details that earn marks

	RAM	ROM
Main use	Running programs and data	Start-up instructions
When power is off	Contents lost (volatile)	Contents retained (non-volatile)
Typical access	Read and write	Mainly read in normal use

01 · THEORY

Virtual memory: the second room

A slower place to hold some data when the worktop (RAM) is full.



01 · THEORY

How virtual memory works

The picture explains why it helps and why it makes some tasks slower.

1

RAM becomes full

A program needs space for more data or instructions.

↓

2

Move out less-needed data

The system transfers some contents from RAM to an area of secondary storage.

↓

3

Keep working

This frees space in RAM for the data currently required.

↓

4

Bring data back when needed

Retrieving it from secondary storage is slower than using RAM.

Do not confuse virtual memory with cache

Cache holds frequently needed data close to the CPU to improve access. Virtual memory uses slower secondary storage to free RAM space when needed.

02 · KEY TERMINOLOGY

Words that earn marks

These precise definitions connect the kitchen pictures to the OCR examination.

Primary storage

Memory that gives the processor access to data or instructions it needs.

RAM

Volatile read/write memory holding currently used programs and data.

ROM

Non-volatile memory used to store computer start-up instructions.

Volatile

Loses its stored contents when the power is switched off.

Non-volatile

Retains stored contents without electrical power.

Firmware

Instructions stored in non-volatile memory for a device to start or operate.

Virtual memory

An area of secondary storage used when available RAM space is insufficient.

Secondary storage

Non-volatile storage, such as an SSD or HDD, generally slower than RAM.

Tip · Use the term, then explain what it does in the question’s context.

03 · WORKED EXAMPLES

Worked Example 1 · RAM vs ROM

A model response with clearly explained reasoning, not a classroom “I Do”.

QUESTION · 4 MARKS

Describe two differences between RAM and ROM.

Step 1 · Choose two genuine contrasts

What they store

RAM stores currently used programs and data. ROM stores start-up instructions.

Whether contents persist

RAM is volatile, so its data is lost without power. ROM is non-volatile and retains its contents.

Step 2 · Write your answer in complete comparisons

Model response: RAM stores programs and data currently being used, while ROM stores the instructions needed to start the computer. RAM is volatile and loses its contents without power, whereas ROM is non-volatile and keeps them.

Why it earns marks: You identify two features of RAM and give the matching contrast for ROM (4 linked marking points).

03 · WORKED EXAMPLES

Worked Example 2 · RAM runs out

Use the second-room analogy, then write the computer-science explanation.

QUESTION · 6 MARKS

Explain the purpose of virtual memory and the effect it has on performance.

Picture	Ingredients not needed immediately move from the crowded worktop to the back room.
Technical process	When RAM is full, some data is moved to an area of secondary storage.
Return	When needed again, data is transferred back into RAM.
Consequence	Secondary storage is slower than RAM, so repeated transfers reduce performance.

Exam-quality six-mark explanation

Virtual memory allows a computer to continue running when available RAM is full. The system transfers some data from RAM to an area of secondary storage, freeing space for other data and programs. When the moved data is needed again, it is brought back into RAM. This is slower because secondary storage takes longer to access than RAM, so frequent swapping can reduce performance.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Answer independently. The mark scheme is on separate pages.

Q1 What is RAM? **1 mark**

Q2 What does volatile mean? **1 mark**

Q3 What is ROM used for? **1 mark**

Q4 State two differences between RAM and ROM. **2 marks**

Q5 State two things stored in RAM. **2 marks**

04 · EXAM PRACTICE

Your turn · Questions 6–8

Answer independently. The mark scheme is on separate pages.

Q6 Explain why computers need RAM. **3 marks**

Q7 Explain why ROM is non-volatile and why that matters. **3 marks**

Q8 Explain what virtual memory is. **3 marks**

04 · EXAM PRACTICE

Your turn · Questions 9–11

Answer independently. The mark scheme is on separate pages.

Q9 Describe two differences between RAM and ROM. **4 marks**

Q10 Describe what happens when a computer uses virtual memory. **4 marks**

Q11 Explain the purpose of virtual memory and the effect it has on performance. **6 marks**

05 · ANSWERS & MARK SCHEME

Mark your answers · Q1–Q6

Mark for the computer-science idea, not just a memorised phrase.

Q1 · 1 mark

RAM is volatile main memory holding programs and data in current use.

Marking: 1: Holds running programs/data (main working memory).

Q2 · 1 mark

Volatile memory loses its stored data when power is switched off.

Marking: 1: Contents are lost without power.

Q3 · 1 mark

ROM stores start-up instructions or firmware used when the computer boots.

Marking: 1: Boot/start-up instructions.

Q4 · 2 marks

RAM is volatile whereas ROM is non-volatile. RAM stores current programs and data whereas ROM contains start-up instructions.

Marking: 1 for each valid contrasted difference (2 total).

Q5 · 2 marks

Running programs and data/files currently in use.

Marking: 1: Programs; 1: Data currently being processed.

Q6 · 3 marks

RAM temporarily holds programs and data while they run, allowing the CPU fast access to instructions and data required for processing.

Marking: 1: Stores active programs; 1: Stores active data/instructions; 1: CPU can access rapidly.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q7–Q11

Use the marking ideas to identify anything you need to revisit.

Q7 · 3 marks

ROM is non-volatile, meaning its contents remain when the computer is turned off. This matters because start-up instructions are still available when power is switched on again.

Marking: Non-volatile (1); retains stored instructions without power (1); allows start-up on the next boot (1).

Q8 · 3 marks

Virtual memory uses an area of secondary storage when RAM does not have enough space. Data is transferred out of RAM into this area until it is needed again.

Marking: Insufficient RAM (1); uses secondary storage (1); temporarily transfers data (1).

Q9 · 4 marks

RAM is volatile and stores currently running programs and data. ROM is non-volatile and typically stores start-up instructions.

Marking: RAM active data/programs (1); ROM start-up instructions (1); RAM volatile (1); ROM non-volatile (1).

Q10 · 4 marks

When RAM is full, the operating system moves some less-needed data from RAM to a reserved area of secondary storage. This makes room in RAM for other data. If the moved data is needed again, it is brought back.

Marking: RAM reaches capacity (1); data moves to secondary storage (1); frees RAM space (1); returns when required (1).

Q11 · 6 marks

Virtual memory enables work to continue when available RAM is full. The computer moves some data from RAM to an area of secondary storage. This frees RAM space for active programs and data. When the moved data is needed, it is returned to RAM. Secondary storage is slower to access than RAM, so these transfers can slow the computer, particularly if frequent.

Marking: Why needed (1); uses secondary storage (1); moves data out of RAM (1); frees space (1); moves it back (1); explains reduced speed (1).

01 · THEORY

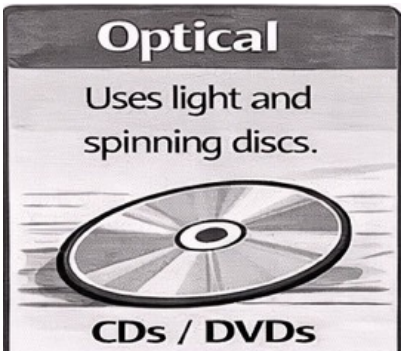
Why we need secondary storage

A device needs non-volatile storage to keep files when power is off. Use the diagram to compare optical, magnetic and solid-state devices.

Big idea · memory versus storage

RAM holds the data currently being used; secondary storage keeps files and programs when power is off. It is non-volatile.

OPTICAL



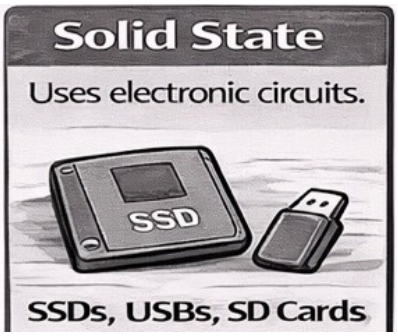
Discs: CD, DVD

MAGNETIC



Hard disk drives

SOLID STATE



SSDs, flash media

From picture to examination language

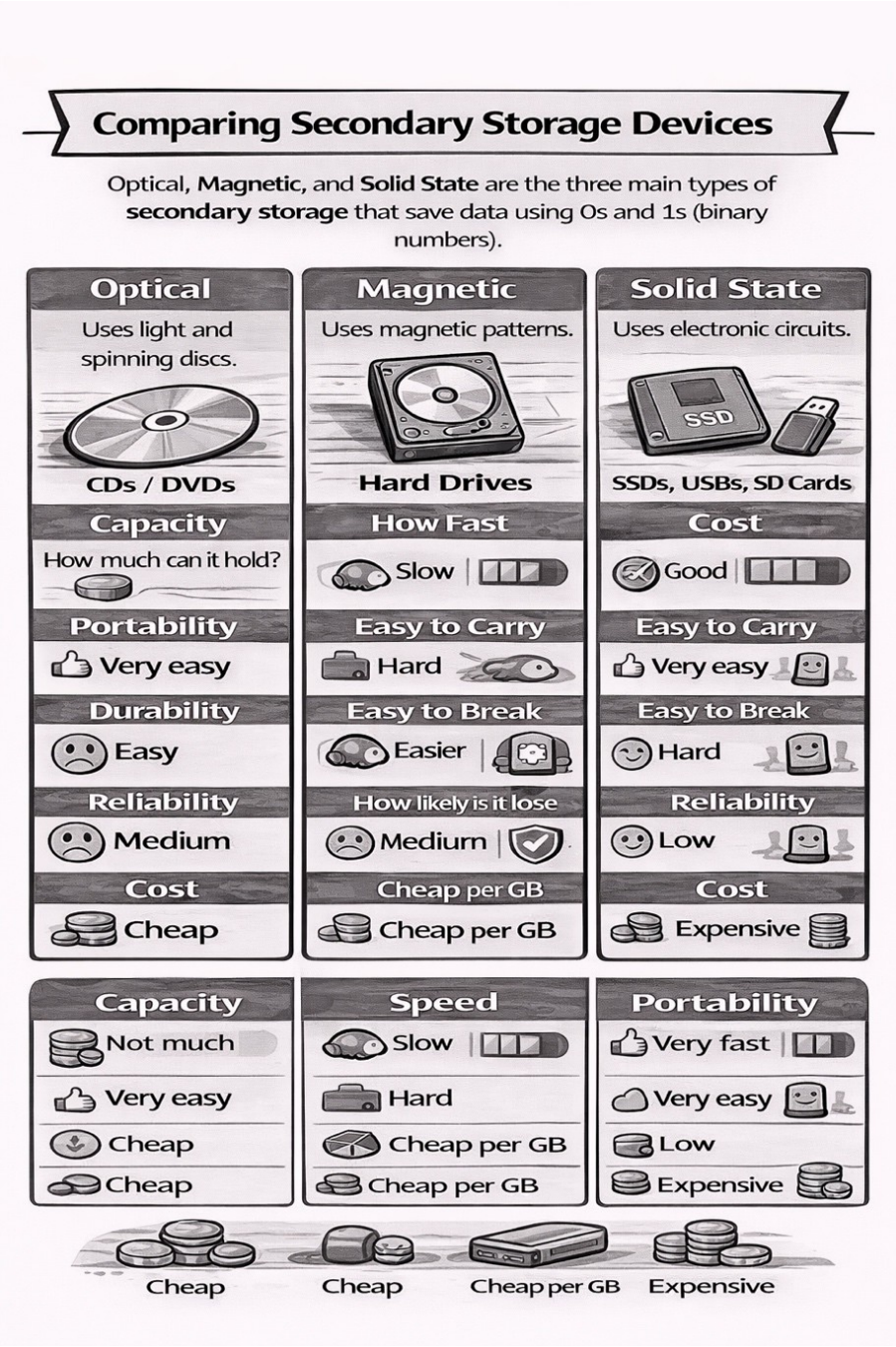
Recognise which type each device uses. When a question asks you to recommend one, justify it against the scenario: capacity, speed, portability, durability, reliability and cost.

Use the picture to recall each storage type. Check capacity, speed, durability, reliability and cost using the comparisons that follow.

01 · THEORY

Compare three types of secondary storage

Use the diagrams to compare optical, magnetic and solid-state storage. The devices shown are examples, not a fixed ranking chart.

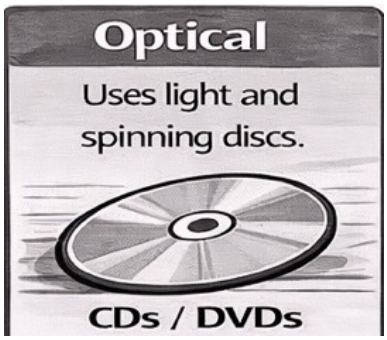


Devices vary in reliability and portability. For exam answers, compare capacity, speed, cost, durability, reliability and portability.

01 · THEORY

Three types, three different methods

The image explains what the storage devices look like; the wording below is what earns marks.



Optical

Examples: CDs, DVDs, Blu-ray discs.

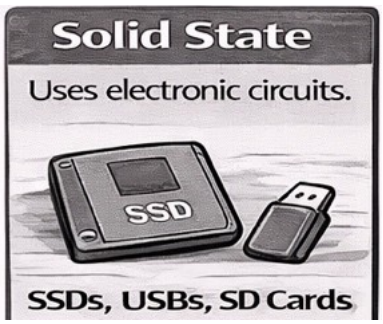
A laser reads the disc. Often convenient for distributing physical media; a disc can be scratched.



Magnetic

Example: hard disk drive (HDD).

A hard disk stores data as magnetic patterns. Typically offers large capacity at a relatively low cost per GB.



Solid state

Examples: SSD, memory card, USB flash drive.

Stores data electronically with no moving parts. Typically gives faster access and better shock resistance.

OCR does not require detailed knowledge of the component parts inside each type of storage device; concentrate on recognition, comparison and choice.

01 · THEORY

Which storage type should you choose?

There is no single best storage device. Compare the requirement with the characteristics that matter in the question.

Capacity	**Total amount stored.** Magnetic HDDs can offer very large capacities; check the particular device.
Speed	**How quickly data is accessed.** SSDs are commonly faster than HDDs; optical discs are often slower.
Portability	**Ease of moving or carrying it.** Flash drives and memory cards are small; optical discs can be carried but scratched.
Durability	**Resistance to physical damage.** SSDs have no moving parts; HDDs have moving mechanisms; discs can scratch.
Reliability	**Likelihood of retaining data and working.** Depends on age, use, environment and failure risk: no type is infallible.
Cost	**Price, often cost per GB.** HDDs often provide lower cost per GB; SSDs trade some price for speed.

Exam technique: Name a type, give a relevant characteristic, then link that characteristic to the user’s needs.

02 · KEY TERMINOLOGY

Words that earn marks

The exact translucent yellow strokes used in J277/H446 lessons sit behind each term.

Secondary storage	Non-volatile storage used for files and programs between sessions.
Non-volatile	Retains its stored data when the power is switched off.
Optical storage	Storage media read using light, such as CDs and DVDs.
Magnetic storage	Stores data using magnetic patterns, such as an HDD.
Solid-state storage	Electronic, non-volatile storage with no moving parts.
Capacity	The maximum quantity of data a device can store.
Access speed	How quickly data can be read or written.
Portability	How easily a device can be moved or carried.
Durability	How well a device withstands physical damage.
Reliability	How dependably data remains available over time.
Cost per GB	Cost relative to the amount of storage provided.

03 · WORKED EXAMPLES

Worked Example 1 · Backing up files

Use the actual storage characteristics, not blanket statements such as “solid state is always best”.

SCENARIO · 4 MARKS

A film club keeps a very large archive of videos at a fixed desk. It wants plenty of storage without spending too much. Recommend a type of secondary storage and justify it.

1 Identify the most important requirements

Very large capacity and a relatively low cost per GB.

2 Choose a suitable type

Magnetic storage, such as a large hard disk drive.

3 Link the type to the scenario

An HDD can provide high capacity economically, making it suitable for large video files.

4 State the sensible limitation

It has moving parts and may be slower than an SSD; portability matters less at a fixed desk.

03 · WORKED EXAMPLES

Worked Example 2 · Portable computing

Same method, different needs. A different choice can earn full marks when the explanation fits.

SCENARIO · 4 MARKS

A pupil takes a laptop to school each day. It must start quickly, survive frequent movement and use little power. Recommend a storage type and explain why.

Recommended type: solid-state storage

An SSD usually gives fast access for loading programs. With no moving parts, it is better suited to being carried than a typical spinning HDD.

Link reasons to the scenario

Fast access supports quick start-up; resistance to knocks supports a laptop that is frequently moved. You are explaining the needs, not just listing features.

Balanced judgement

For maximum capacity at minimum cost per GB, an HDD might still be sensible. The best answer depends on what the question values most.

Marking approach: award clear choice + justified points linked to the scenario. More than one type can sometimes be defended.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Marks are shown for each question.

Q1 What is secondary storage? — 1 mark

Q2 Give one example of secondary storage. — 1 mark

Q3 What does non-volatile mean? — 1 mark

Q4 State two reasons computers need secondary storage. — 2 marks

Q5 State two differences between primary and secondary storage. — 2 marks

04 · EXAM PRACTICE

Your turn · Questions 6–9

Marks are shown for each question.

Q6 **Explain how optical storage stores data.** **3 marks**

Q7 **Explain how magnetic storage stores data.** **3 marks**

Q8 **Explain how solid-state storage stores data.** **3 marks**

Q9 **Describe two advantages of solid-state storage.** **4 marks**

04 · EXAM PRACTICE

Your turn · Questions 10–11

Marks are shown for each question.

Q10 **Describe two disadvantages of optical storage.** **4 marks**

Q11 **Compare optical, magnetic and solid-state storage.** **6 marks**

05 · ANSWERS & MARK SCHEME

Mark your answers · Q1–Q5

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q1 **What is secondary storage?** **1 mark**

Secondary storage is non-volatile storage used to store data and programs permanently.

Q2 **Give one example of secondary storage.** **1 mark**

Hard disk drive / SSD / USB flash drive / CD / DVD.

Q3 **What does non-volatile mean?** **1 mark**

Non-volatile means data is kept when the power is switched off.

Q4 **State two reasons computers need secondary storage.** **2 marks**

It stores files permanently and keeps data when the computer is turned off.

Q5 **State two differences between primary and secondary storage.** **2 marks**

Compared with RAM, secondary storage is non-volatile and typically slower to access. ROM is a non-volatile form of primary storage.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q6–Q9

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q6 **Explain how optical storage stores data.** **3 marks**

Optical storage uses laser light to read data stored as pits and lands on a spinning disc.

Q7 **Explain how magnetic storage stores data.** **3 marks**

Magnetic storage stores data using magnetic patterns on a spinning disc that can be read and written by a magnetic head.

Q8 **Explain how solid-state storage stores data.** **3 marks**

Solid-state storage stores data using electronic circuits and has no moving parts.

Q9 **Describe two advantages of solid-state storage.** **4 marks**

Solid-state storage offers fast access, so files and programs load quickly. It has no moving parts, making it more resistant to damage from knocks when carried.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q10–Q11

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q10 **Describe two disadvantages of optical storage.** **4 marks**

Optical discs usually hold less data than large HDDs/SSDs, so are unsuitable for very large backups. Access can be slower, so reading a large collection of files takes longer.

Q11 **Compare optical, magnetic and solid-state storage.** **6 marks**

Optical discs are portable and can be cheap per disc, but typically offer lower capacity and slower access. Magnetic HDDs can hold large amounts at low cost per GB, but moving parts can be damaged by knocks. Solid-state storage provides fast access and good shock resistance, but may cost more per GB. Choose according to the scenario.

Six-mark comparison checklist

Award credit for accurate, distinct comparisons between all three types. Include the six OCR factors when relevant: capacity, speed, portability, durability, reliability and cost. Avoid fixed claims that one type always wins.

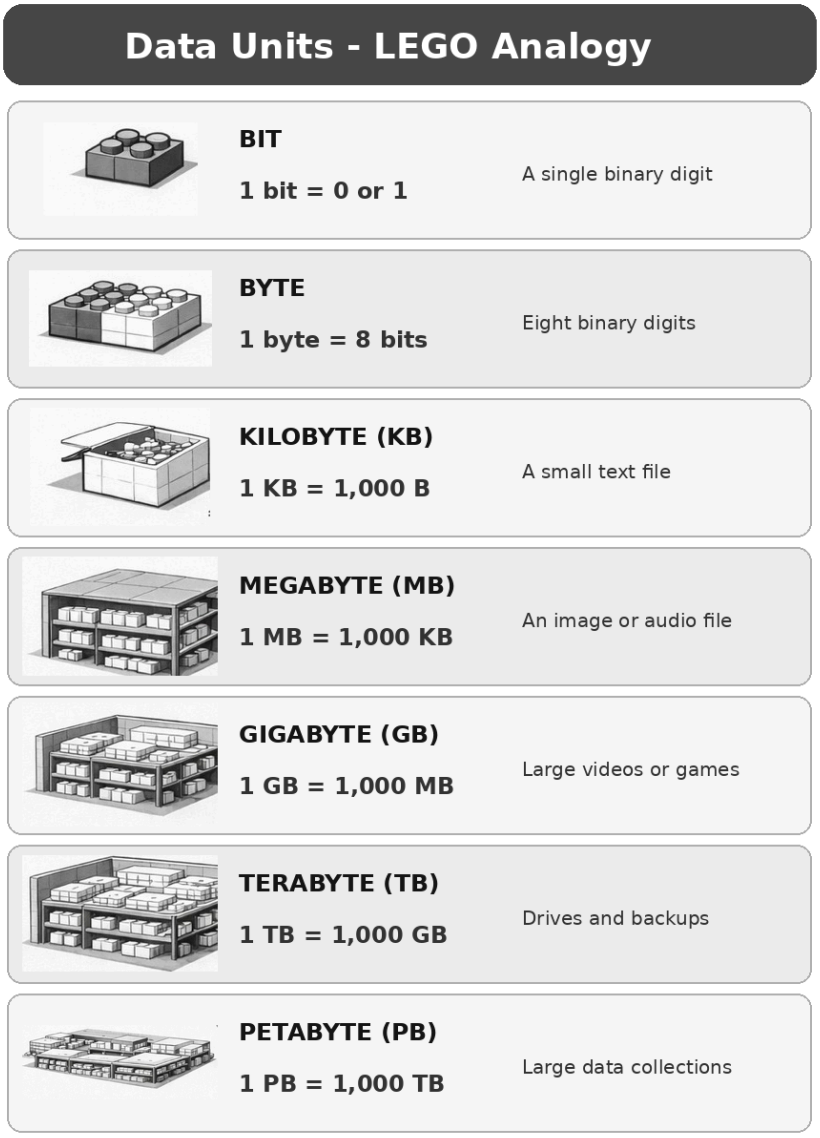
Self-mark strictly

For a 4-mark “describe two” answer, give two different advantages or disadvantages and develop each. For a 6-mark comparison, explain trade-offs: do not simply list three device names.

01 · THEORY

Data units are like LEGO

Start with a single binary digit, then group pieces to make larger units. The LEGO illustration makes the increase in storage units easier to remember.



From bytes onwards: 1,000 of each unit makes the next larger unit.
Bit to byte is different: eight bits make one byte.

From picture to examination language

A bit is one 0 or 1. 4 bits make a nibble; 8 bits make a byte. From bytes, OCR normally uses decimal steps of 1,000.

Use the LEGO blocks to picture the increasing size of each unit. The next page shows the exact conversions.

01 · THEORY

How big is each unit?

The LEGO stack grows; the table gives the exact units and conversions you should use in OCR answers.

Bit	**b**	A single 0 or 1
Nibble	**4 bits**	Half a byte
Byte	**8 bits**	Two nibbles
Kilobyte (KB)	**1,000 B**	1,000 bytes
Megabyte (MB)	**1,000 KB**	One thousand kilobytes
Gigabyte (GB)	**1,000 MB**	One thousand megabytes
Terabyte (TB)	**1,000 GB**	One thousand gigabytes
Petabyte (PB)	**1,000 TB**	One thousand terabytes

Move between units

Small → large: divide by 1,000 per step. Large → small: multiply by 1,000 per step. Bits ↔ bytes use 8, not 1,000.

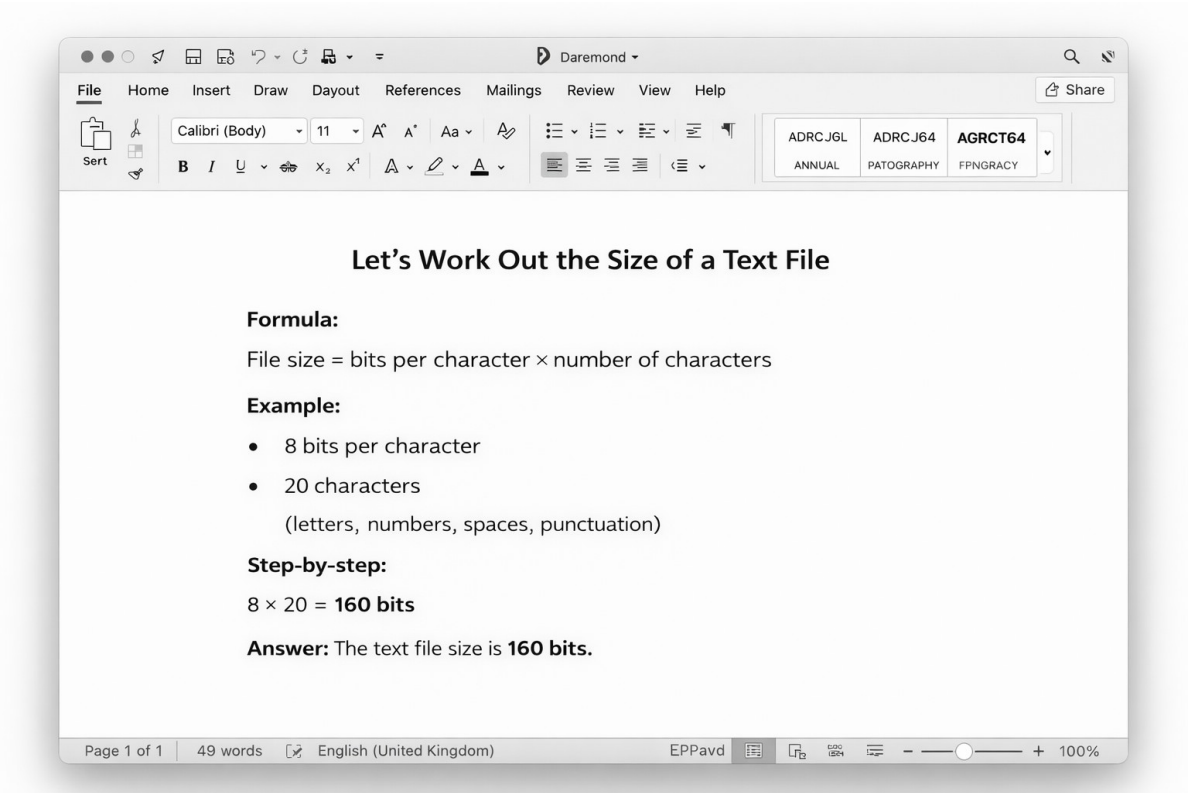
Capacity is not free space

Capacity is the fixed maximum for that device; free space changes as files are added or removed.

01 · THEORY

A text file is made of bits

The illustrated text-file calculation shows how a character count becomes a size in bits. It is an example, not a rule that every character is 8 bits.



Text file size formula

Number of characters × bits per character = file size in bits. Then divide by eight for bytes. A question may state an 8-bit character representation.

Worked example · character file size

20 characters × 8 bits = 160 bits. Divide by eight: 20 bytes. The number of bits per character must be provided or stated as an assumption.

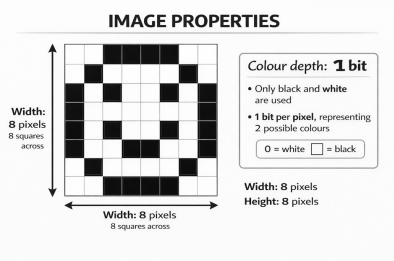
A character set assigns codes to letters and symbols. Unicode provides a much wider character repertoire than ASCII. Do not assume every Unicode character requires 16 or 32 bits.

01 · THEORY

Picture size: count pixels

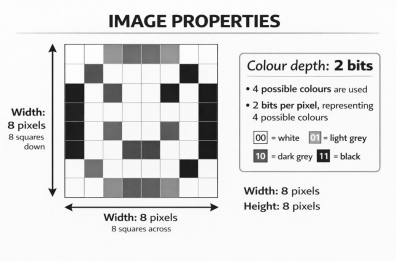
The pixel-grid images show that file size depends on image dimensions and the number of bits used for each pixel.

1-bit colour



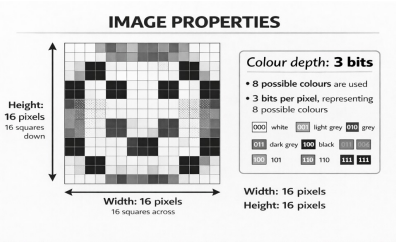
8 × 8 × 1 = 64 bits

2-bit colour



8 × 8 × 2 = 128 bits

3-bit colour



16 × 16 × 3 = 768 bits

Image file size (uncompressed)

Width (pixels) × height (pixels) × colour depth (bits per pixel) gives the number of bits. Divide by 8 for bytes.

How to work out a question

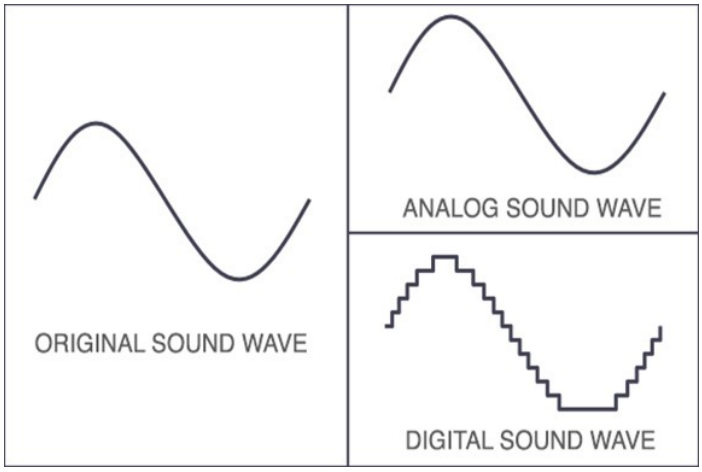
Write the formula; multiply dimensions to get pixels; multiply by colour depth; then convert bits → bytes → KB → MB if needed.

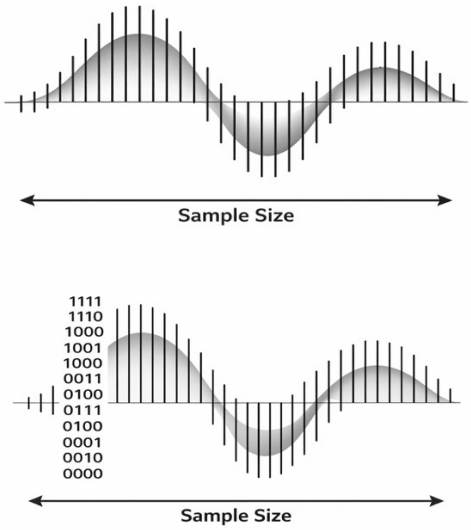
Exam reminder: this is the uncompressed pixel-data size and normally excludes file headers and metadata unless the question includes them.

01 · THEORY

Sound size: store each sample

The waveform diagrams show how an analogue sound is sampled and represented as binary data.





Sound file size (uncompressed, one channel)

Sample rate (samples/s) × duration (seconds) × bit depth (bits/sample) gives a file size in bits. Then divide by eight for bytes.

Sample rate	More measurements each second → larger file.
Duration	More seconds of recording → larger file.
Bit depth	More bits representing each sample → larger file.

Use the correct exam terms: sample rate is samples per second and bit depth is bits per sample.

02 · KEY TERMINOLOGY

Keywords · Part 1

Precise terms for OCR written answers, with the same three yellow highlighter pen strokes as the J277 lessons.

Bit	One binary digit, either 0 or 1.
Nibble	A group of four bits.
Byte	Eight bits.
Kilobyte	1,000 bytes using the OCR decimal convention.
Megabyte	1,000 kilobytes.
Gigabyte	1,000 megabytes.
Terabyte	1,000 gigabytes.
Petabyte	1,000 terabytes.

Check your units: bits vs bytes use 8; larger decimal units use 1,000.

02 · KEY TERMINOLOGY

Keywords · Part 2

Precise terms for OCR written answers, with the same three yellow highlighter pen strokes as the J277 lessons.

Capacity	Maximum quantity of data a device can store.
Free space	Unused portion of the current capacity.
Character set	Mapping between characters and their coded values.
Bits per character	Storage bits used for each character as given in a question.
Colour depth	Number of bits used to represent each pixel.
Sample rate	Number of audio samples taken each second.
Bit depth	Number of bits used to represent each audio sample.
Duration	How long the recording lasts, measured in seconds.

Check your units: bits vs bytes use 8; larger decimal units use 1,000.

03 · WORKED EXAMPLES

Worked Example 1 · LEGO to files

Follow each conversion rather than guessing whether to multiply or divide.

A · Convert 4,500 KB into MB

4,500 ÷ 1,000 = 4.5 MB. The unit is getting larger, so the number becomes smaller.

B · Convert 24 bits into bytes

24 ÷ 8 = 3 bytes. Use eight when moving from bits to bytes; do not divide by 1,000.

C · Calculate text storage

A 30-character file uses 8 bits per character. 30 × 8 = 240 bits. Then 240 ÷ 8 = 30 bytes. State the assumed character size.

The marking method

1. Write the relationship. 2. Substitute the numbers. 3. Calculate. 4. Give the right unit.

These are worked examples. Independent exam questions follow on separate pages, without model answers beside them.

03 · WORKED EXAMPLES

Worked Example 2 · Image data

The exam tells you the dimensions and colour depth; you must convert the result into the requested unit.

Question: An uncompressed image is 400 × 300 pixels with 4-bit colour depth. Calculate its size in KB (using 1,000 bytes per KB).

1. Count pixels

400 × 300 = 120,000 pixels

2. Find the number of bits

120,000 × 4 = 480,000 bits

3. Convert to bytes

480,000 ÷ 8 = 60,000 bytes

4. Convert to KB

60,000 ÷ 1,000 = 60 KB

Exam accuracy

The formula gives uncompressed pixel data. Compression or metadata can change the total if a question includes them.

03 · WORKED EXAMPLES

Worked Example 3 · Sound data

Use the sample rate, bit depth and duration in the single-channel uncompressed-audio formula.

Question: A mono sound recording uses 8,000 samples per second for 5 seconds, at 8 bits per sample. Calculate uncompressed size in KB.

1. Write the formula

Sample rate × duration × bit depth

2. Substitute the values

8,000 × 5 × 8 = 320,000 bits

3. Convert bits to bytes

320,000 ÷ 8 = 40,000 bytes

4. Convert bytes to KB

40,000 ÷ 1,000 = 40 KB

The bigger-file link

Higher sample rate, greater bit depth or longer duration increases the uncompressed data size if other factors stay fixed.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Each question shows its mark allocation.

Q1 What is a bit? — 1 mark

Q2 How many bits are in a byte? — 1 mark

Q3 What is meant by capacity? — 1 mark

Q4 State two differences between capacity and free space. — 2 marks

Q5 State two differences between ASCII and Unicode. — 2 marks

04 · EXAM PRACTICE

Your turn · Questions 6–9

Each question shows its mark allocation.

Q6 **Explain why computers store data in binary.** **3 marks**

Q7 **Explain what a character set is.** **3 marks**

Q8 **Explain how to calculate the size of a text file.** **3 marks**

Q9 **Calculate the size of a text file with 200 characters using ASCII.** **4 marks**

04 · EXAM PRACTICE

Your turn · Questions 10–11

Each question shows its mark allocation.

Q10 **Calculate the size of an image that is 1000×800 at 8-bit colour depth (in MB).** **4 marks**

Q11 **Explain how sample rate, bit depth and duration affect the size of a sound file.** **6 marks**

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 1–5

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q1 **What is a bit?** **1 mark**

A bit is one binary digit: 0 or 1.

Q2 **How many bits are in a byte?** **1 mark**

There are 8 bits in a byte.

Q3 **What is meant by capacity?** **1 mark**

Capacity is the maximum amount of data a storage device can hold.

Q4 **State two differences between capacity and free space.** **2 marks**

Capacity is the maximum amount a device can hold and is unchanged simply by deleting files; free space is the unused part of that capacity and changes as data is added or removed.

Q5 **State two differences between ASCII and Unicode.** **2 marks**

Unicode is able to represent a much wider range of characters and scripts than ASCII. The number of bits per character depends on the encoding; use the value specified in any file-size question.

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 6–8

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q6 **Explain why computers store data in binary.** **3 marks**

Electronic circuits can reliably distinguish two states, such as on/off or high/low voltage (1). Binary represents these using 0 and 1 (1), allowing data to be stored and processed as electronic states (1).

Q7 **Explain what a character set is.** **3 marks**

A character set is an agreed set of characters (1) where each character is associated with a code or numerical value (1), so letters and symbols can be represented in binary (1).

Q8 **Explain how to calculate the size of a text file.** **3 marks**

Multiply the number of characters by the number of bits per character (1). The product is a size in bits (1). Divide by 8 to get bytes, then convert to larger units as needed (1).

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 9–10

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q9 **Calculate the size of a text file with 200 characters using ASCII.** **4 marks**

Assume 8 bits per character, as in the Practice question (1). 200 x 8 = 1,600 bits (1). 1,600 / 8 = 200 bytes (1). Answer: 200 bytes (1).

Q10 **Calculate the size of an image that is 1000×800 at 8-bit colour depth (in MB).** **4 marks**

1,000 x 800 = 800,000 pixels (1). 800,000 x 8 = 6,400,000 bits (1). Divide by 8 = 800,000 bytes (1). Divide by 1,000 twice = 0.8 MB (1).

05 · ANSWERS & MARK SCHEME

Self-mark · Question 11

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q11 **Explain how sample rate, bit depth and duration affect the size of a sound file.** **6 marks**

A higher sample rate means more samples each second, raising the total bits stored (2). Higher bit depth means more bits in each sample, raising the file size (2). Longer duration means more seconds of samples, raising the file size (2). The calculation is sample rate x duration (seconds) x bit depth, giving bits for one channel.

6-mark reasoning checklist

Sample rate: samples each second. Bit depth: bits per sample. Duration: seconds. Explain why increasing each raises uncompressed data size.

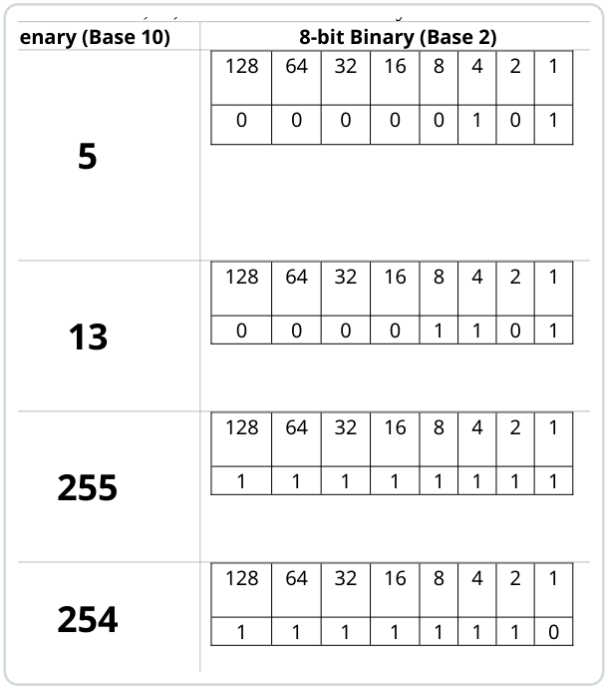
THEORY

Data Storage

Use the place-value diagrams to learn the conversion methods before attempting exam questions.

How to use this section

Study the picture first, follow the worked method, then use the exact terms. The last pages contain **all 15 Practice questions followed by separate mark-scheme answers.**



The eight-bit place-value number line

Three representations, one number

Binary uses base 2; denary uses base 10; hexadecimal uses base 16. The same number can be written in all three formats. For example, **254 = 11111110 (binary) = FE (hex)**.

The full topic also covers binary addition/overflow, shifts, character sets, image pixels/metadata and digital sound. These are all part of OCR J277 1.2.4.

THEORY

Binary and the number line

Use the 128–64–32–16–8–4–2–1 place values.

enary (Base 10)	8-bit Binary (Base 2)							
	128	64	32	16	8	4	2	1
5	0	0	0	0	0	1	0	1
13	0	0	0	0	1	1	0	1
255	1	1	1	1	1	1	1	1
254	1	1	1	1	1	1	1	0

Use the place values: write 1 for a value used, otherwise write 0.

Example: 13 in 8 bits

128	64	32	16	8	4	2	1

8 + 4 + 1 = 13. The highest place (128) is the most significant bit; the smallest place (1) is the least significant bit.

Check the binary value

Check the place values: 1111110 = 126 (64+32+16+8+4+2), whereas 254 is **11111110 in eight bits. Always check the number of bits required.**

WORKED EXAMPLES

Convert to and from binary

Example 1 · 47 denary to binary

Work from left to right: 32 fits (15 left), then 8 (7 left), 4 (3 left), 2 (1 left) and 1. The unused weights become zero.

8-bit place-value number line

128	64	32	16	8	4	2	1

47 = 00101111. Each 1 means that place value is included in the total.

Example 2 · 10000001 binary to denary

8-bit place-value number line

128	64	32	16	8	4	2	1

Only the 128 and 1 columns contain a 1: **128 + 1 = 129**. You can use a shorter number line for fewer than eight digits, for example **1101 = 13**.

Exam wording

Say **place value**, **most significant bit (MSB)** and **least significant bit (LSB)**. Denary whole numbers tested here range from 0 to 255.

THEORY

Hexadecimal: two groups of four

Use the 8?4?2?1 nibble method to link binary, hexadecimal and colour codes.

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	0
8	**4**	**2**	**1**	**8**	**4**	**2**	**1**
8+4+2+1 =15				8+4+2=14			
F				**E**			

128	64	32	16	8	4	2	1
1	0	1	0	0	0	1	1
8	**4**	**2**	**1**	**8**	**4**	**2**	**1**
8+2 =10				2+1=3			
A				**3**			

Split 11111110 into 1111 (F) and 1110 (E).

Hexadecimal uses digits **0–9** then **A=10, B=11, C=12, D=13, E=14, F=15**. Every four binary bits make one hexadecimal digit (a **nibble**).

Remember the colour-code analogy

A six-digit RGB colour such as **#00FF00** uses hexadecimal pairs for the red, green and blue channels: 00 red, FF green, 00 blue. Hex is a shorter notation for binary data, not a replacement for it.

WORKED EXAMPLES

Converting hexadecimal

Example 1 · 254 denary to hexadecimal

Convert 254 to eight bits

128	64	32	16	8	4	2	1

Split into two groups: **1111 | 1110**. The first nibble is 15 (F), the second is 14 (E). So **254 = FE**.

Example 2 · A4 hexadecimal to denary

A = 1010 and 4 = 0100

128	64	32	16	8	4	2	1

A=10 and 4=4, so the binary number is 10100100. Add weights with 1s: **128 + 32 + 4 = 164**. Alternatively **10 × 16 + 4 = 164**.

Check your method

Convert each nibble carefully, then check your answer using denary place values.

THEORY

Binary addition and overflow

Use column addition, carry bits and the eight-bit limit to check each result.

Example 1 · add 13 and 5 in eight bits

```
 00001101   (13)
+ 00000101   (5)
----------
 00010010   (18)
```

Add from the right. When 1+1 occurs, write **0** in that column and carry **1** to the next, just as carrying works in denary.

Example 2 · overflow

```
 11110000   (240)
+ 00110000   (48)
----------
1 00100000   (288 total)
```

An 8-bit register can only hold eight result bits, representing **0–255** for an unsigned number. The ninth carry bit cannot fit: this is an **overflow error**. Keeping only the low eight bits gives **00100000 = 32**, which is not the true sum.

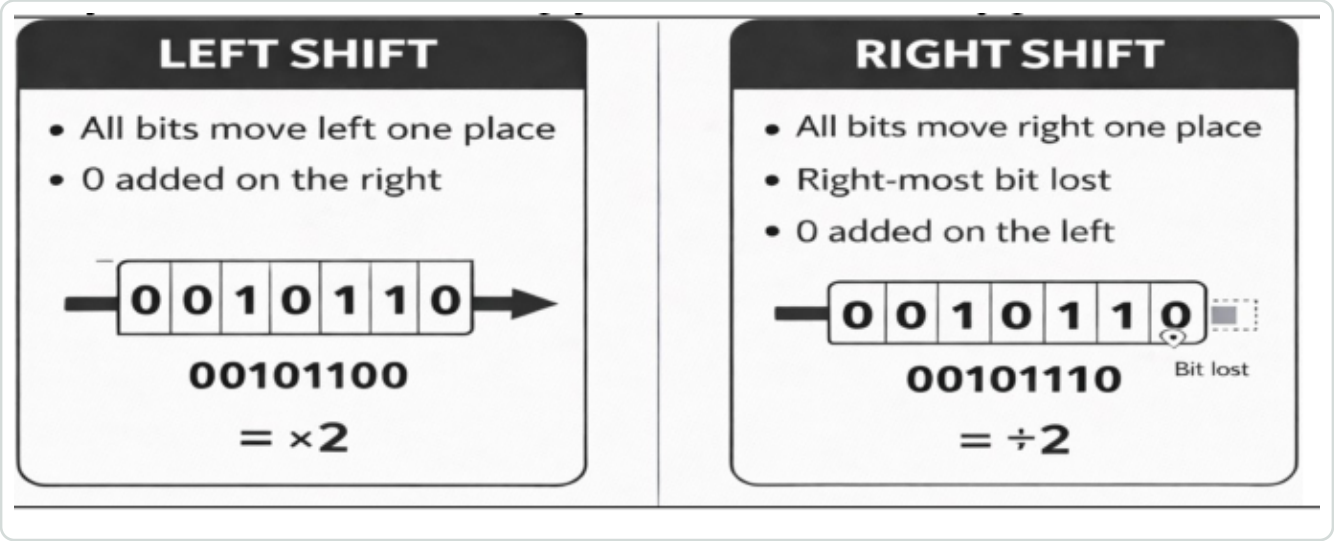
Worked model wording

An overflow error occurs when the result is too large for the available number of bits. The carry outside the eight-bit range is not stored in the eight-bit result.

THEORY

Binary shifts

Shifting binary digits changes their place values.



The empty positions created by a logical shift are filled with zero.

Before: 00010110 = 22

128	64	32	16	8	4	2	1

After a two-place left shift = 88

128	64	32	16	8	4	2	1

When no significant bits are lost, a two-place **left shift multiplies by 4**. A right shift of two places divides an unsigned value by four, discarding any bits shifted off the right edge.

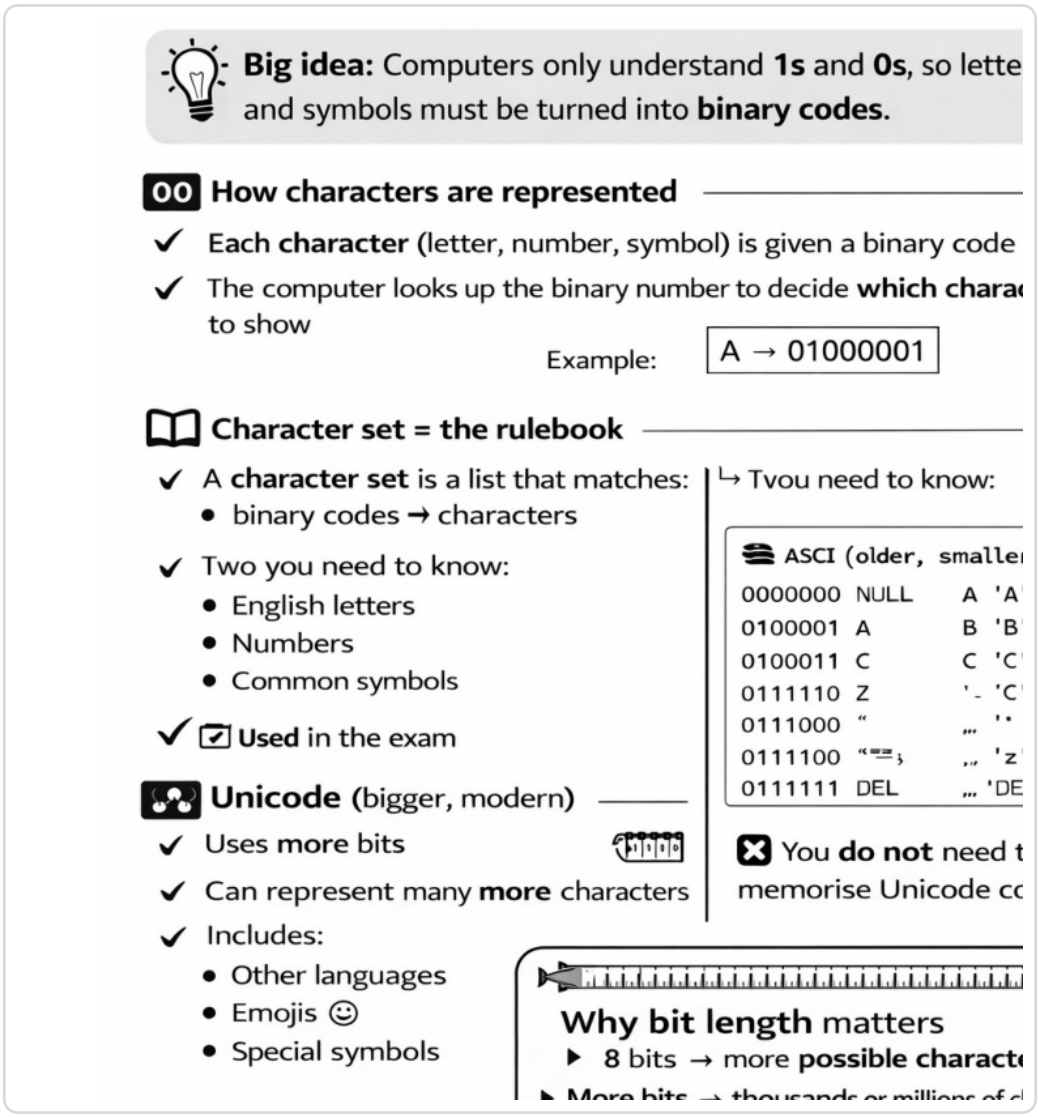
Important limit

Bits shifted beyond a fixed 8-bit boundary are **lost**. For example, 11000000 (192) shifted left one place becomes 10000000 (128) in eight bits, because the leftmost bit is discarded. Do not always claim shifting left doubles the value.

THEORY

Character sets: ASCII and Unicode

Use the annotated diagram to understand character encoding.



Use the character-set diagram to compare ASCII and Unicode.

A **character set** assigns numerical codes to characters. OCR gives eight-bit binary ASCII codes in questions; students do not need to memorise the character codes. The sequence is ordered: B has a code one greater than A.

Make the distinction precise

Unicode supports far more characters and languages than ASCII; a Unicode **encoding** can use different numbers of bytes. Avoid claiming every Unicode character always uses 16 or 32 bits.

WORKED EXAMPLES

Character codes follow a sequence

Worked example · the next three letters after C

Start with C = 01000011 in eight-bit ASCII. Find the codes for D, E and F without memorising them. Each next character increases the number by one.

C · 01000011

128	64	32	16	8	4	2	1

D · 01000100

128	64	32	16	8	4	2	1

E · 01000101

128	64	32	16	8	4	2	1

F · 01000110

128	64	32	16	8	4	2	1

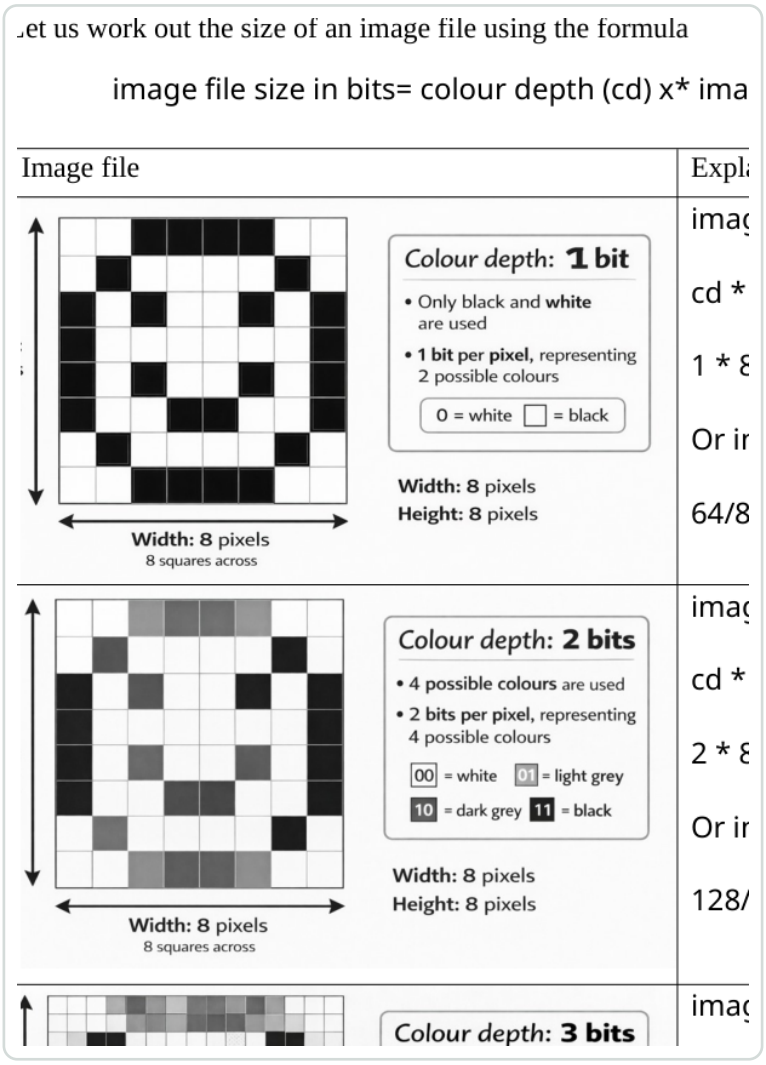
Exam wording

Explain that character codes are **ordered** and move forward by one for successive letters in this example. A character set determines how stored binary patterns are interpreted as characters.

THEORY

Images: pixels and colour depth

Use the pixel grids to understand resolution and colour depth.



These grids show pixels represented using different bit depths.

An image consists of **pixels**. Each pixel stores a binary colour code. **Resolution** tells you how many pixels the image contains; **colour depth** tells you how many bits are used to store each pixel's colour.

Increasing resolution can capture more spatial detail; increasing colour depth can represent more colours. Both generally increase uncompressed image data size.

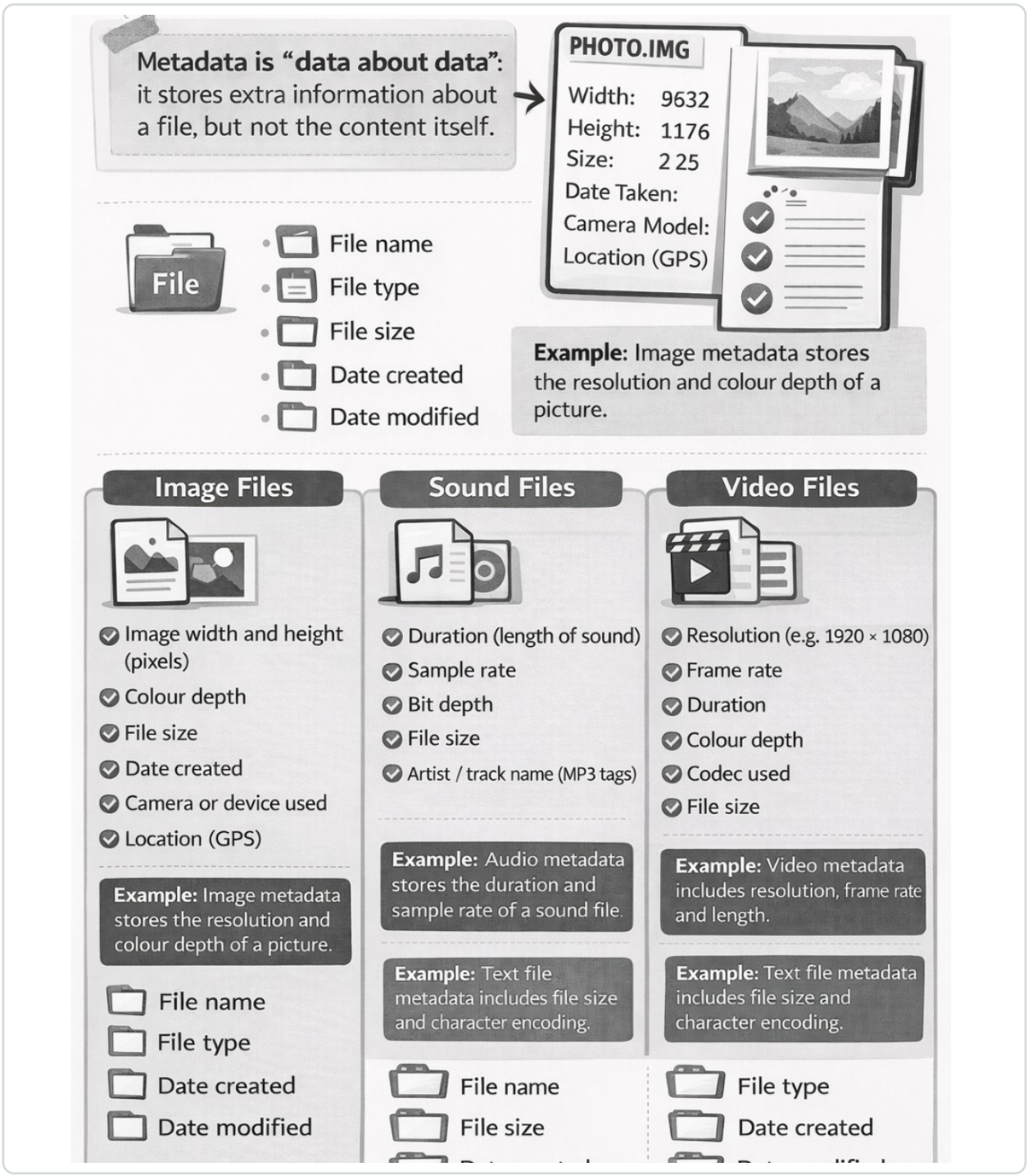
Synoptic link to 1.2.3

Uncompressed image data size in bits = width × height × colour depth. This calculation was introduced in the preceding units topic and is useful for explaining changes in file size.

THEORY

Metadata is data about the file

Use the examples to identify metadata stored with files.



Metadata can describe image, audio and video file properties.

Examples include image width and height, capture date, GPS location or file type. **Metadata describes the content or file; it is not the image pixels themselves.**

THEORY

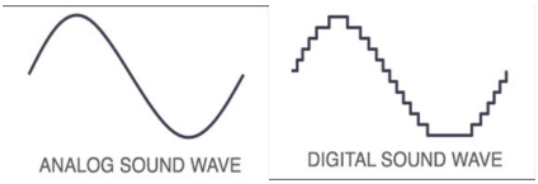
Sound becomes binary by sampling

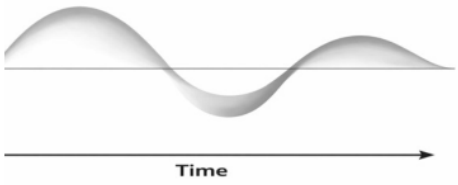
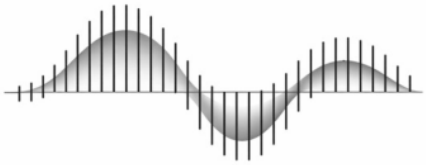
Compare the analogue and sampled sound waveforms.

	10,000 ÷ 1000 = **10 megabytes**

Analogue sound is smooth and continuous, with no steps, and is how sound exists in the real world.

Digital sound is how computers store sound by taking many snapshots (samples) and saving them as binary 0s and 1s) because computers cannot store sound/analogue waves.



Term	Definition	Image
Duration	How long the sound lasts (time along the bottom) Measured in **seconds**	
Sample rate	How often the sound is measured each second (how close the vertical sample lines are) Measured in **Hertz**	

These waveform diagrams help distinguish samples and bit depth.

Sample rate is measurements per second, measured in hertz (Hz). **Bit depth** is the number of bits stored for each amplitude sample. **Duration** is how long the recording lasts.

More samples per second or more possible amplitude levels can improve the digital representation, but they also increase raw file size. Longer recordings store more samples.

Exam-accurate units

Bit depth is measured in **bits per sample**, not “the number of 1s and 0s used to represent height.” Sample rate is measured in **Hz**, not bits.

KEY TERMINOLOGY

Numbers and operations

Binary	Base-2 number system using 0 and 1.
Denary	Base-10 number system using digits 0 to 9.
Hexadecimal	Base-16 number system using 0–9 and A–F.
Bit	One binary digit, either 0 or 1.
Nibble	A group of four bits; equivalent to one hex digit.
MSB	Most significant bit: the highest place value.
LSB	Least significant bit: the lowest place value.
Binary shift	Moves bits left or right within a fixed width.
Overflow	A result outside the number range available in the given bits.
Character set	An agreed mapping between numbers and characters.

KEY TERMINOLOGY

Characters, images and sound

ASCII	Character-code standard; OCR represents ASCII as eight bits in exam questions.
Unicode	Standard supporting characters from many writing systems.
Pixel	A picture element storing a colour code.
Resolution	Width and height of an image in pixels.
Colour depth	Number of bits used for each pixel’s colour.
Metadata	Information about the file, such as dimensions or capture date.
Sampling	Measuring an analogue sound signal at regular time intervals.
Sample rate	Number of amplitude samples taken each second, in Hz.
Bit depth	Number of bits available to store each audio sample.
Duration	Length of a recording, in seconds.

WORKED EXAMPLES

Put the methods together

Example 1 · mixed binary and hex question

Convert **01101101** into denary and hexadecimal. Add the weights where the bits are 1: 64+32+8+4+1 = **109**. Split 0110 | 1101 to get 6 and D: **6D**.

8-bit place-value number line

128	64	32	16	8	4	2	1

Example 2 · shifting a binary integer

Shift **00010110** two places left, keeping eight bits. Move every bit two places left and fill the two empty spaces on the right with 0s: **01011000**.

8-bit place-value number line

128	64	32	16	8	4	2	1

Its value changes from 22 to **88**, which is four times as large because no significant bits were lost.

WORKED EXAMPLES

Explain representation changes

Images · changing colour depth

A small bitmap is **8 × 8 pixels**. At 1 bit per pixel, its raw pixel data is 64 bits. At 2 bits per pixel it needs 128 bits. Double the colour depth → double the raw pixel-data size if the pixel dimensions stay the same.

Sound · changing sample rate

A mono clip lasting five seconds sampled at 10 Hz and 8-bit depth stores **10 × 5 × 8 = 400 bits** of raw samples. Doubling only the sample rate to 20 Hz doubles the raw size to **800 bits**.

Character sets · counting available codes

With 8 bits, there are **2^8 = 256** unique binary patterns. An encoding using more available patterns can represent a larger collection of distinct symbols. The actual byte count per encoded character depends on the encoding.

Practice first, answers later

Attempt all 15 questions. Keep the answers separate until you have finished.

EXAM PRACTICE

Practice questions

Try all 15 questions (43 marks). Answers follow the practice section.

Q1 What is binary? **1 mark**

Q2 What does hexadecimal use instead of only numbers? **1 mark**

Q3 What is a binary shift? **1 mark**

Q4 State two reasons hexadecimal is used. **2 marks**

Q5 State two rules of binary shifts. **2 marks**

Q6 Explain how to convert a denary number into binary. **3 marks**

EXAM PRACTICE

Questions — continued

Answers are in a separate section. Complete each question before turning to the mark scheme.

Q7 **3 marks**

Explain how to convert binary to denary.

Q8 **3 marks**

Explain the effect of a left binary shift by 2 places.

Q9 **3 marks**

Explain what happens to bits that move off the end in a binary shift.

Q10 **3 marks**

Explain what metadata is and give one example of metadata stored in a file.

Q11 **3 marks**

Explain why hexadecimal does not replace binary.

EXAM PRACTICE

Questions — continued

Answers are in a separate section. Complete each question before turning to the mark scheme.

Q12 **4 marks**

Convert 13 into 8-bit binary.

Q13 **4 marks**

Convert the hexadecimal number A4 into denary.

Q14 **4 marks**

The ASCII code for C is 01000011. Work out the ASCII codes for D, E and F.

Q15 **6 marks**

Explain how denary numbers are converted into hexadecimal.

EXAM PRACTICE

Additional OCR coverage

These five additional questions practise specification content not covered by the preceding 15 questions.

QE1 **3 marks**

Add 00001101 and 00000101 using eight-bit binary addition. Show the carry when it is needed.

QE2 **3 marks**

An eight-bit register adds 11110000 and 00110000. Explain why an overflow error occurs, and state the eight-bit result.

QE3 **3 marks**

Explain what the terms MSB and LSB mean when interpreting 10100001.

QE4 **4 marks**

An image doubles in both width and height but keeps its colour depth. Explain the effects on the number of pixels, potential detail, and uncompressed size.

QE5 **3 marks**

Explain the difference between sample rate and bit depth in digital sound.

ANSWERS & MARK SCHEME

Practice questions: model answers

Indicative marking points for the 15 practice questions.

Q1 · 1 mark

Binary is a **base-2** number system using only **0** and **1** (1).

Q2 · 1 mark

It uses the letters **A–F** to represent values **10–15** (1).

Q3 · 1 mark

A binary shift moves the bits in a binary number left or right by a specified number of places (1).

Q4 · 2 marks

Two points: hexadecimal represents a long binary pattern with fewer symbols (1), and is easier for people to read, remember or communicate (1).

Q5 · 2 marks

An unsigned left shift by one **doubles** a value if no significant bits are lost (1). A right shift by one **halves** the integer value, discarding any remainder (1).

Q6 · 3 marks

Use the **128–64–32–16–8–4–2–1 place values** (1). Choose which weights sum to the denary number (1). Write 1 under selected weights, otherwise 0, keeping eight places where required (1).

Q7 · 3 marks

Identify the binary place values (1). Select only those positions whose bit is 1 (1). Add their weights to obtain the denary value (1).

Q8 · 3 marks

Every bit moves two places to the left (1), zero bits fill the empty right-hand positions (1), and the unsigned value is multiplied by 4 **only if no significant bits are lost** (1).

ANSWERS & MARK SCHEME

Answers — continued

Use these indicative marking points to self-mark after attempting the questions.

Q9 · 3 marks

A bit shifted past the fixed-width boundary is **discarded** (1), it is absent from the stored result (1), and the information cannot be recovered from that result (1).

Q10 · 3 marks

Metadata is information **about** the image/file rather than its pixel/audio contents (1). Give a plausible item, e.g. image width or height (1); explain that it helps software interpret or display the file (1).

Q11 · 3 marks

Computers process and store patterns of binary bits (1). Hexadecimal is a shorter way for people to write those bit patterns (1). Each hex digit maps exactly onto four bits (1).

Q12 · 4 marks

Write the eight place values (1). Use 8, 4 and 1 (1). Fill the other places with zeros (1). Final eight-bit answer **00001101** (1).

Q13 · 4 marks

A = 10 (1). Convert A and 4 to nibbles **1010 0100** (1). Use place values or 10×16 +4 (1). Final answer: **164** denary (1).

Q14 · 4 marks

The letter codes follow an ordered sequence (1). **D = 01000100** (1), **E = 01000101** (1), and **F = 01000110** (1).

Q15 · 6 marks

Identify the denary input and use an eight-bit binary representation (1). Select 128–1 place values correctly (1). Split the eight bits into two four-bit groups (1). Work out each nibble’s value using 8–4–2–1 (1). Convert 10–15 into hex letters A–F where needed (1). Present the resulting two-digit hexadecimal number in the correct order (1).

ANSWERS & MARK SCHEME

Additional OCR coverage

These answers apply only to the five additional questions.

QE1 · 3 marks

Show correct alignment (1), binary carrying within the rightmost columns (1), and the result **00010010 (18)** (1).

QE2 · 3 marks

240+48 = 288 (1). This exceeds 255, the unsigned eight-bit maximum, producing a ninth carry bit (1). Only eight result bits fit: **00100000** (1).

QE3 · 3 marks

MSB is the leftmost bit/highest place value (1); LSB is the rightmost/lowest place value (1); in 10100001 their weights are 128 and 1 respectively (1).

QE4 · 4 marks

Doubling width doubles each row of pixels (1), doubling height doubles the number of rows (1), the total pixel count therefore quadruples (1), and with unchanged colour depth, the uncompressed pixel data also quadruples (1).

QE5 · 3 marks

Sample rate is the number of amplitude measurements taken per second and is measured in Hz (1). Bit depth is the number of bits used for each amplitude measurement (1). Raising either can increase raw file size and potentially representation quality, but they affect different parts of the recording (1).

Exam reminder: Use accurate technical definitions alongside visual analogies.
Check units and worked calculations against the correct method on these pages.
Show your working wherever the question requires it.

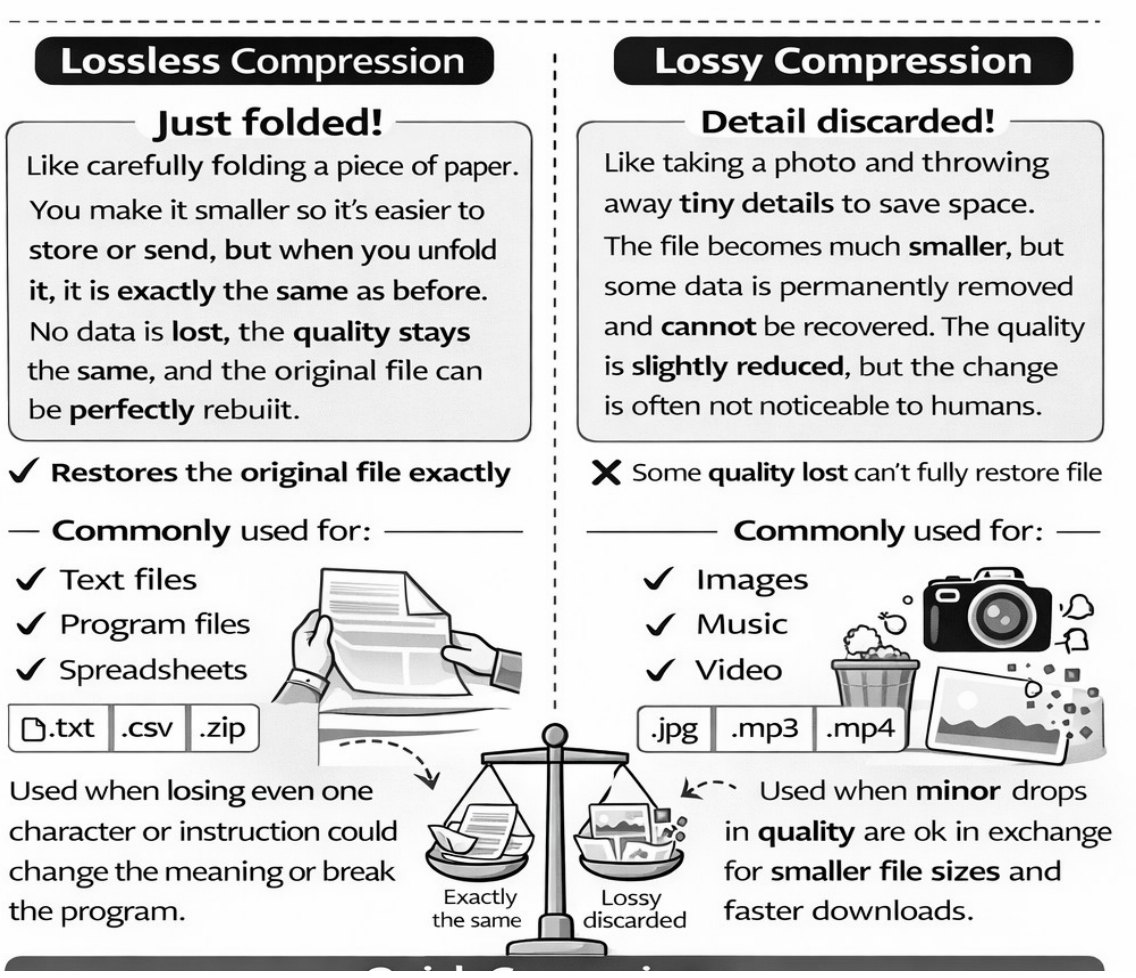
THEORY

Compression: a smaller file

Begin with the visual analogy, then use the technical explanation in an OCR exam answer.

THE BIG IDEA

Compression reduces the amount of data needed to represent a file. This can save storage space and reduce how much data must be transmitted.



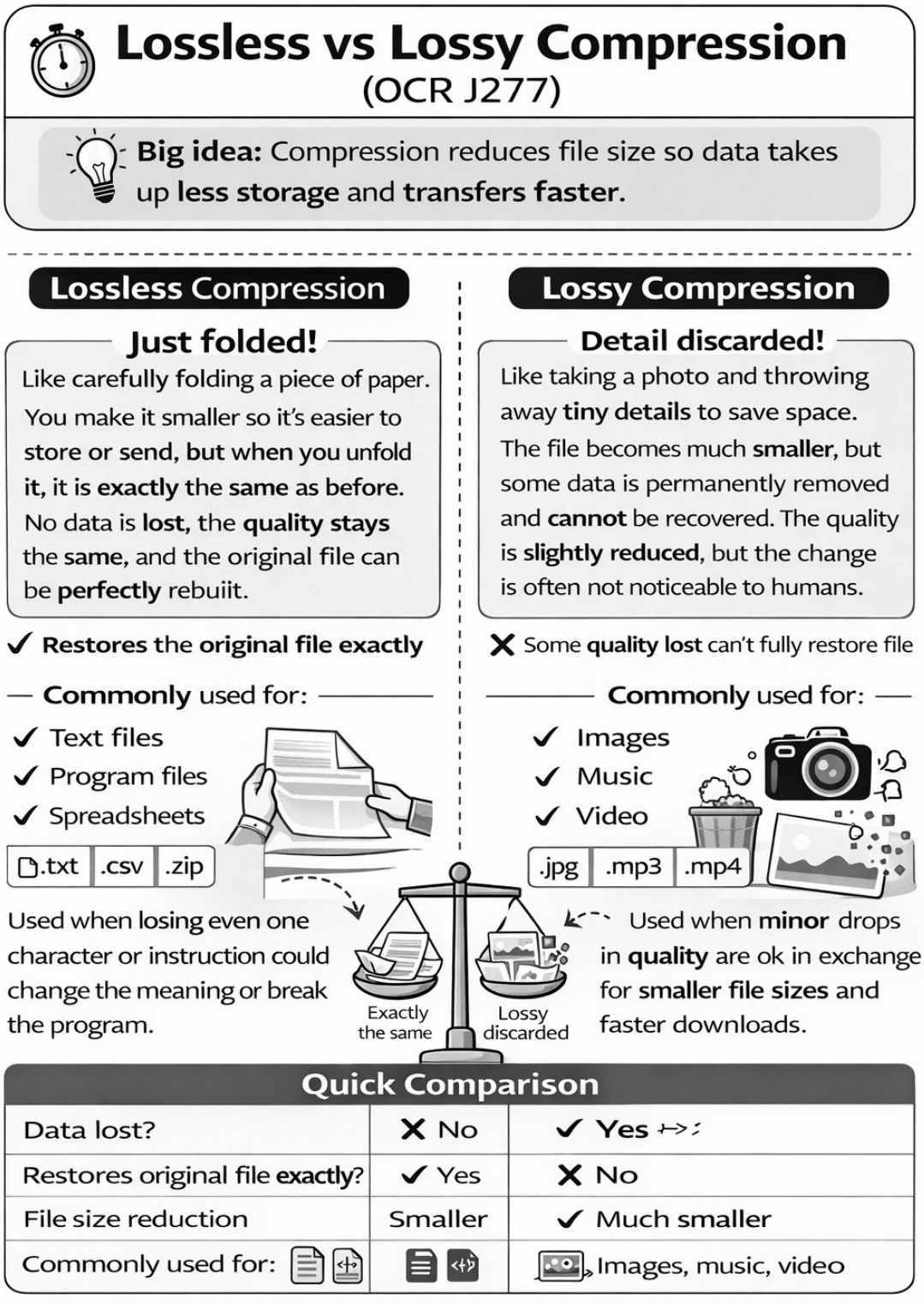
Lossless: like folding paper; recover the original exactly.
Lossy: like throwing away some detail; smaller file, but exact recovery is impossible.

Study the picture, then compare the two compression methods.

THEORY

Compression explained with an analogy

The folded-paper and photograph examples show lossless and lossy compression.



THEORY

What the analogy tells us

Start with the picture, then use the correct technical words in your explanation.

1 LOSSLESS / THE FOLDED DOCUMENT

The original information is encoded more efficiently. When decompressed, every bit of the original data can be reconstructed. That makes it appropriate for source code, text documents and other files where missing information would matter.

Exam wording: lossless compression reduces file size while allowing exact restoration, with no original data lost.

2 LOSSY / THE PHOTO WITH DETAIL REMOVED

Some information is discarded permanently, often details which are less noticeable in photographs, sound or video. This can produce smaller files, but the original cannot be restored exactly. Visible or audible quality can be reduced.

Exam wording: lossy compression permanently removes some data, trading the ability to restore the original exactly for reduced file size.

CHECK THE DIFFERENCE BETWEEN LOSSY AND LOSSLESS

Lossy can achieve greater size reductions, but the amount saved varies with the content, format, method and settings. Do not memorise “lossless = small” and “lossy = always much smaller” as universal rules.

THEORY

How to choose a compression type

The best method depends on whether the content must be recovered exactly and how much quality can be sacrificed.

Question to ask	Lossless	Lossy
Can all original data be recovered?	Yes	No
What happens to data?	No permanent loss	Some data discarded
What happens to quality?	Original preserved	May be reduced
When is it useful?	Code, documents, archives	Photos, music, video
Is file size reduction fixed?	No; depends on file	No; depends on file

WHY COMPRESS?

Smaller files require less storage and usually less data to send across a network. Transfer time can therefore fall when connection conditions are otherwise similar. Compression and decompression can also require processing.

KEY TERMINOLOGY

Words to use in exam answers

The highlighted words use the same three translucent SVG pen strokes as the J277 and H446 lessons.

Compression	Reducing the size of a file or the amount of data needed to represent it.
Lossless	Compression that allows the original information to be reconstructed exactly.
Lossy	Compression that permanently discards some information.
Decompression	Reconstructing data from its compressed representation.
File size	The amount of storage space used by a file.
Storage capacity	The total amount of data a storage device can hold.
Transfer time	Time taken to send or receive a file.
Quality	How faithfully media represents the original image or sound.
Exact restoration	Reconstructing all of the original data without loss.

WORKED EXAMPLES

Worked example 1 · School files

Scenario: a school is compressing examination answers and its programming coursework.

DECIDE

Should the school choose lossless or lossy compression? Explain two reasons for your decision.

STEP 1 IDENTIFY THE REQUIREMENT

Text and source code must be reproduced exactly. Even one changed character could alter a sentence or a program.

STEP 2 CHOOSE THE COMPRESSION TYPE

Select lossless compression, because no original information is permanently discarded.

STEP 3 LINK TO THE BENEFIT

The files take up less storage and can be transferred using less data, while still being restored exactly.

Model answer: Use lossless compression because the original examination and program files must be recovered exactly. It reduces file size without removing data that could change the answers or break code.

WORKED EXAMPLES

Worked example 2 · Website photos

Scenario: a school website contains many photographs and must load efficiently on mobile connections.

DECIDE

Explain why lossy compression might be chosen for the published photographs, and give one disadvantage.

STEP 1 IDENTIFY THE NEED

Many large images use storage and must be transferred to visitors. Smaller image files can help the pages load faster.

STEP 2 CHOOSE THE TRADE-OFF

Lossy compression may remove image detail that viewers are unlikely to notice, reducing the transmitted file size.

STEP 3 STATE THE DISADVANTAGE

Some original image information is permanently removed. High compression settings may cause visible artefacts; the original cannot be reconstructed exactly.

Model answer: Lossy compression makes web images smaller so less data needs transferring. However, some visual information is permanently discarded and image quality may be reduced. The school should retain the original separately if needed.

EXAM PRACTICE Practice questions · Q1?Q6

Answer these independently. The solutions and indicative marking points are on later pages.

Q1 What is compression? **1 mark**

Q2 What is lossless compression? **1 mark**

Q3 What is lossy compression? **1 mark**

Q4 State two benefits of compression. **2 marks**

Q5 State two differences between lossless and lossy compression. **2 marks**

Q6 Explain why lossless compression is used for text files. **3 marks**

EXAM PRACTICE Practice questions · Q7?Q10

Explain each decision using the compression type, what happens to the data and the practical benefit.

Q7 Explain why lossy compression is suitable for images, sound, or video. **3 marks**

Q8 Explain how compression improves data transfer. **3 marks**

Q9 Describe the difference between lossless and lossy compression. **4 marks**

Q10 Describe two situations where lossy compression would be used. **4 marks**

EXAM PRACTICE Practice question · Q11

This is a 6-mark explain/discuss response. Plan advantages, disadvantages and a concluding judgement before writing.

Q11 Explain the advantages and disadvantages of lossy compression. **6 marks**

SUGGESTED ANSWER STRUCTURE (NOT A MODEL ANSWER)

1. Explain how lossy compression reduces data and file size.
2. Link smaller files to a practical storage or transfer benefit.
3. Describe permanent data loss or possible quality reduction.
4. Say when that trade-off is acceptable.

ANSWERS & MARK SCHEME

Model answers · Q1–Q6

Use these after attempting the questions. Marking points are indicative; OCR mark schemes may accept equivalent wording.

Q1 · 1 MARK

What is compression?

Model answer: Compression reduces the size of a file so it takes up less storage space.

Q2 · 1 MARK

What is lossless compression?

Model answer: Lossless compression reduces file size without losing any data.

Q3 · 1 MARK

What is lossy compression?

Model answer: Lossy compression reduces file size by permanently removing some data.

Q4 · 2 MARKS

State two benefits of compression.

Model answer: Compression reduces storage space needed and allows files to be transferred faster.

Q5 · 2 MARKS

State two differences between lossless and lossy compression.

Model answer: Lossless compression does not lose data and can be restored exactly, while lossy compression loses data and cannot be restored exactly.

Q6 · 3 MARKS

Explain why lossless compression is used for text files.

Model answer: Text files must be restored exactly, as losing data could change words or instructions.

ANSWERS & MARK SCHEME

Model answers · Q7–Q11

Give yourself credit for correct equivalent points, not only these exact sentences.

Q7 · 3 MARKS

Explain why lossy compression is suitable for images, sound, or video.

Model answer: Some data can be removed without noticeable loss of quality, greatly reducing file size.

Q8 · 3 MARKS

Explain how compression improves data transfer.

Model answer: Smaller files require less data to be sent, so they transfer more quickly over a network.

Q9 · 4 MARKS

Describe the difference between lossless and lossy compression.

Model answer: Lossless compression reduces file size without losing data and allows exact restoration, while lossy compression permanently removes some data and cannot be restored exactly.

Q10 · 4 MARKS

Describe two situations where lossy compression would be used.

Model answer: Lossy compression is used for images and video where a small reduction in quality is acceptable.

Q11 · 6 MARKS

Explain the advantages and disadvantages of lossy compression.

Model answer: Lossy compression greatly reduces file size, improving storage and transfer speeds. However, data is permanently lost and quality may be reduced, meaning the original file cannot be restored exactly.